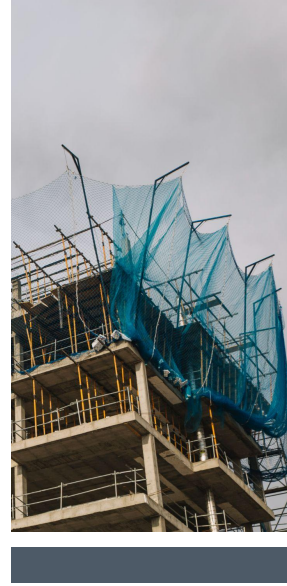
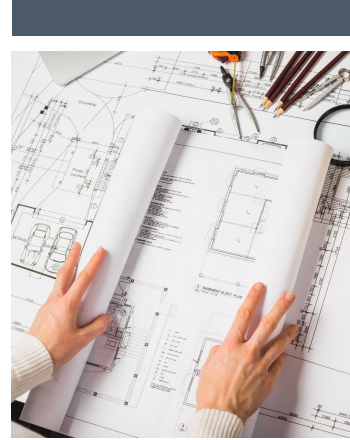


Agile Software Engineering

Software Engineering Course – 2
2024-2025



Agenda

01

Requirements

02

User Personas

03

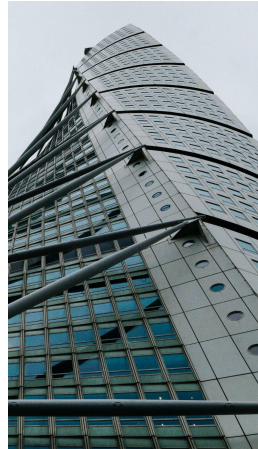
Features

04

Product Backlog

05

**Managing technical
risk**



Requirements Gathering

Informal Requirements
Gathering

Personal Experience

Empirical Evidence

Business Analysis

Formal Requirements
Gathering

Survey

Market Research

Business Analysis

Agile

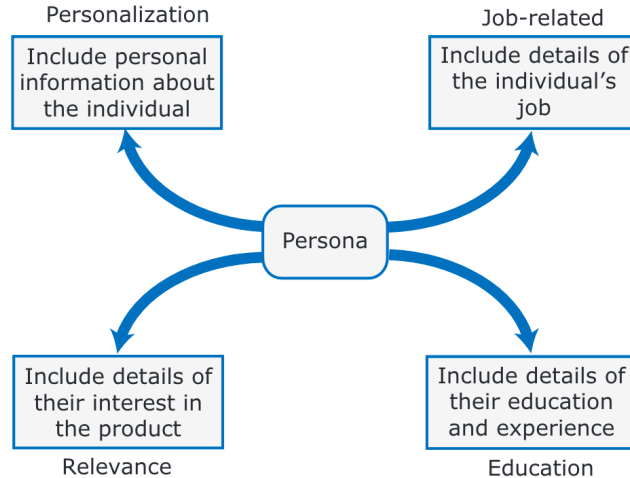


FACEMASH

User Personas

Who are we building this for?

Figure 3.4 Persona descriptions



User Personas

Who are we building this for?

Figure 3.4 Persona descriptions

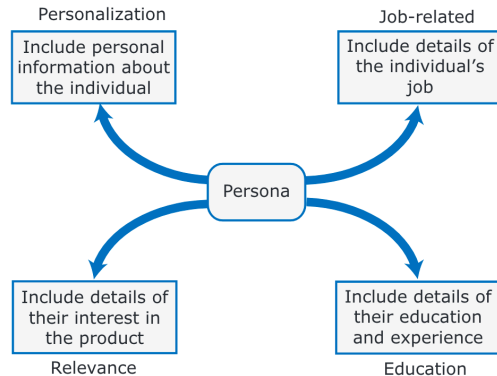


Table 3.4 A persona for an IT technician

Elena, a school IT technician

Elena, age 28, is a senior IT technician in a large secondary school (high school) in Glasgow with over 2000 students. Originally from Poland, she has a diploma in electronics from Potsdam University. She moved to Scotland in 2011 after being unemployed for a year after graduation. She has a Scottish partner, no children, and hopes to develop her career in Scotland. She was originally appointed as a junior technician but was promoted, in 2014, to a senior post responsible for all the school computers.

Although not involved directly in teaching, Elena is often called on to help in computer science classes. She is a competent Python programmer and is a “power user” of digital technologies. She has a long-term career goal of becoming a technical expert in digital learning technologies and being involved in their development. She wants to become an expert in the iLearn system and sees it as an experimental platform for supporting new uses for digital learning.

Derive scenarios from User Personas (ESP p.69 - 75)

Anatomy of a User Story



As a *<type of user>* I want *<some feature, capability of the system>* so that *<I achieve some goal>*.

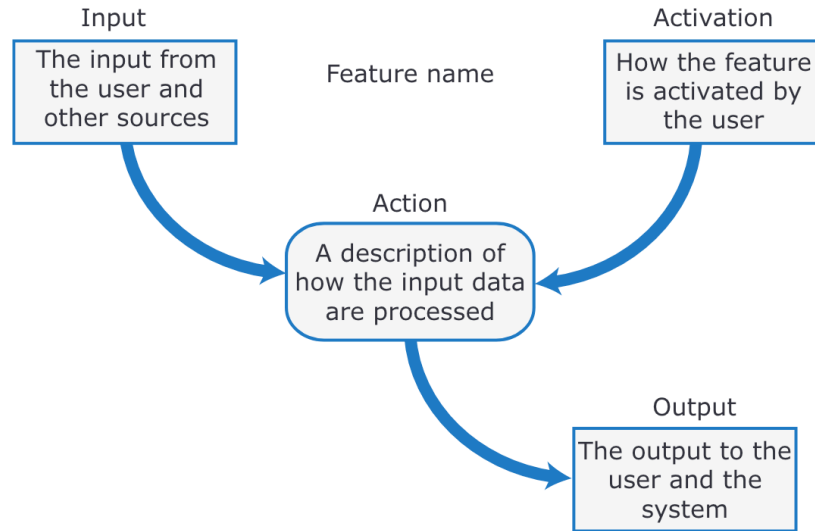
A specific type of user with a specific, identifiable job

A capability, function or feature that is required for the system to support the user

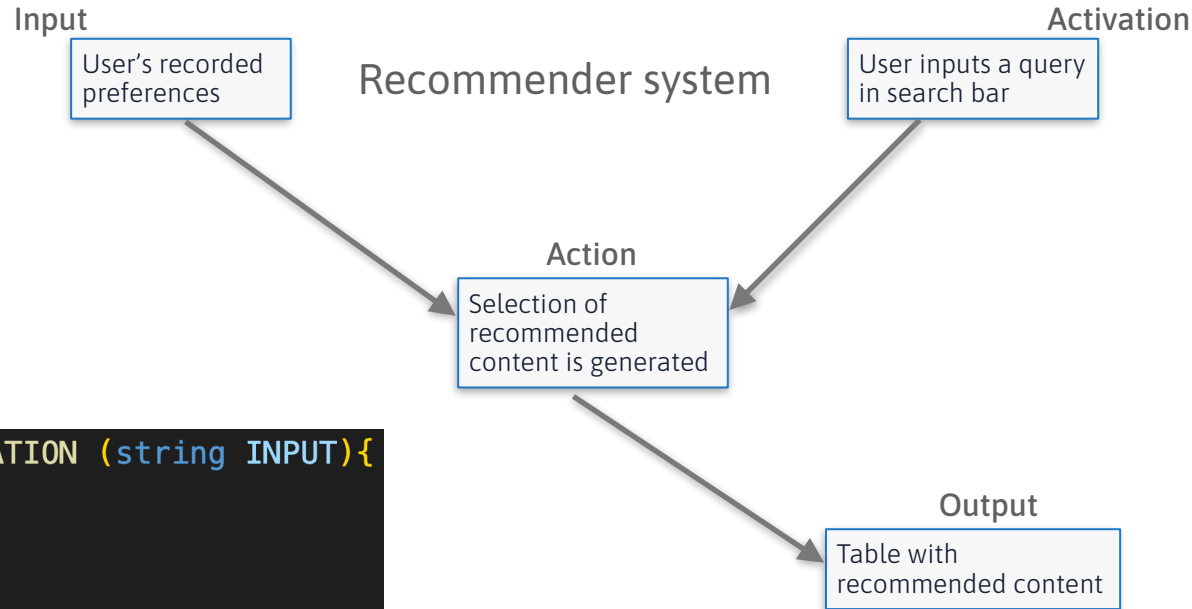
The goal the user is looking to achieve, or the value they are looking to derive using the system

Thinking in Features

Figure 3.2 Feature description

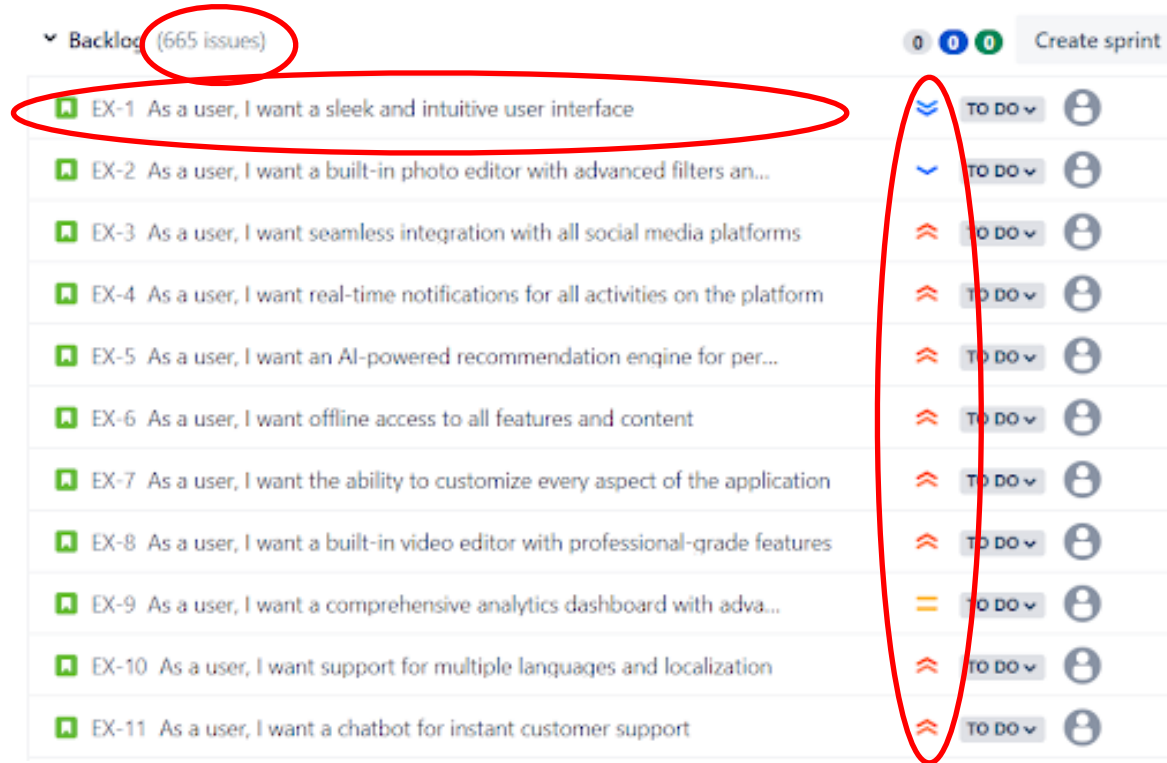


Thinking in Features



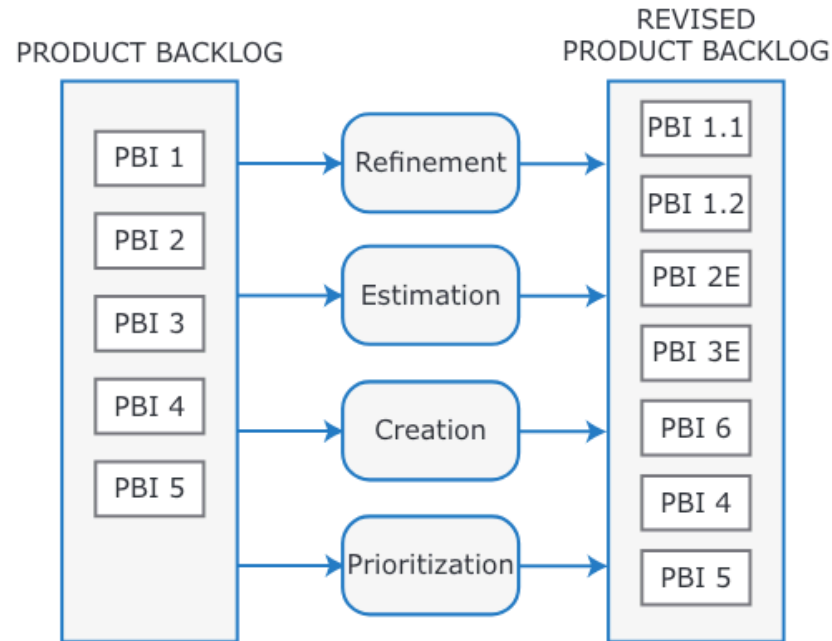
```
function ACTIVATION (string INPUT){  
    //...  
    //ACTION  
    //...  
    💡 return OUTPUT;  
}
```


Product Backlog



<https://theproductmanager.com/topics/product-backlog-examples/>

Product Backlog



Product Backlog

▼ Sprint 1 Add dates (3 issues)

0 0 0 Start sprint

EX-666 As a new user, I want to sign up for a Spotify account. TO DO

EX-667 As a user, I want to log in to my Spotify account. TO DO

EX-668 As a user, I want to search for songs and artists. TO DO

+ Create issue

▼ Sprint 2 Add dates (3 issues)

0 0 0 Start sprint

EX-669 As a user, I want to create and manage playlists. TO DO

EX-670 As a user, I want to play songs from my library. TO DO

EX-671 As a user, I want to discover new music based on my preferences. TO DO

+ Create issue

▼ Near Future Add dates (3 issues)

0 0 0 Start sprint

EX-672 As a user, I want to follow and unfollow artists. TO DO

EX-673 As a user, I want to view and edit my profile information. TO DO

EX-674 As a user, I want to share songs or playlists with my friends. TO DO

+ Create issue

▼ Backlog (21 issues)

0 0 0 Create sprint

EX-675 As a user, I want to like and comment on songs. TO DO

EX-676 As a user, I want to explore curated playlists and recommendations. TO DO

EX-677 As a user, I want to create and manage collaborative playlists. TO DO

EX-678 As a user, I want to personalize my home screen with favorite artists or art... TO DO

Product Backlog

Backlog 10 issues

Create sprint



- | | | | | |
|--|-------|------------------------------|--------|---|
|  As a user, I want to have a spell-checking feature that highlights and corrects misspelled words in the Web Editor | 1.0.0 | Basic Text Editing | EX2-5 | - |
|  As a user, I want Grammarly to provide advanced grammar suggestions in the Web Editor | 1.0.0 | Advanced Editing | EX2-6 | - |
|  As a user, I want Grammarly to detect potential instances of plagiarism in the Web Editor | 1.1.0 | Advanced Editing | EX2-7 | - |
|  As a user, I want Grammarly to provide suggestions for improving the tone of my writing in the Web Editor | 1.2.0 | Advanced Editing | EX2-8 | - |
|  As a user, I want Grammarly to ensure consistency in my writing by suggesting synonyms in the Web Editor | 1.2.0 | Advanced Editing | EX2-9 | - |
|  As a user, I want Grammarly to analyze the readability of my text in the Web Editor | 1.3.0 | Readability Analysis | EX2-10 | - |
|  As a user, I want Grammarly to help me improve the structure of my writing in the Web Editor | 1.3.0 | Readability Analysis | EX2-11 | - |
|  As a user, I want Grammarly to suggest alternative word choices in the Web Editor | 1.3.0 | Readability Analysis | EX2-12 | - |
|  As a user, I want Grammarly to provide real-time writing assistance in the Browser Extension | 1.0.0 | Real-time Writing Assistance | EX2-13 | - |

Product Backlog

User Story: As a user, I want to sign up for a new account.

Requirements:

- Implement a registration form with the following fields:
 - Username: Allow the user to enter a unique username for their account.
 - Email: Provide a field for the user to input their email address.
 - Password: Include a password field for the user to set their account password.
- Implement password validation rules to ensure security:
 - Minimum password length of 8 characters.
 - Requirement for a combination of uppercase and lowercase letters, numbers, and special characters.
- Email Verification:
 - Integrate an email verification process for account activation.
 - Send an email to the provided email address with a verification link.
 - Upon clicking the verification link, verify the email and activate the user's account.
- Handle Error Cases:
 - Implement error handling for scenarios such as duplicate usernames or invalid email addresses.
 - Display appropriate error messages to the user and prompt them to correct the information.

User Story: As a user, I want to integrate my images with Social Media Platforms.

Requirements: We want the user to be able to directly post images to social media. We're currently looking at integrating with Instagram and Twitter.

Product Backlog



Product Backlog Template

This is a Hustle Badger template. Get more practical how-to guides, templates and case studies [HERE](#).

How to use this sheet: make a copy for your private use & edit there - so that others can continue to use this master

OBJECTIVE	Increase annual recurring revenue (ARR) by 1% (\$1.2m) by 30th April 2023									
Name	Theme	Description	PRD	Jira	Every win	Reach	Impact	Confidence	Effort	
Google sign in	Account creation	Allow users to sign up with Google account	Notion	Jira	23-Mar	40%	\$740,776	30%	5	
"Free" -> "Basic"	Pricing options	Reposition "free" membership as "basic" m	Notion	Jira	25-Mar	100%	\$90,000	30%	2	
Magic link	Account creation	Allow users to sign up with email address c	Notion	Jira	05-Apr	60%	\$250,000	30%	8	
Monthly payment	Pricing options	Allow users to purchase paid membership	Notion	Jira	25-Apr	100%	\$150,000	30%	14	
Trust signals	Check out	Add trust signals to check out	Notion	Jira	Early May	100%	\$123,000	5%	3	
Streamline check out	Check out	Remove unnecessary fields from check ou	Notion	Jira	Early May	100%	\$123,000	5%	3	
Rate limit basic	Pricing options	Allow basic members to access one lessor	Notion	Jira	Late May	100%	\$123,000	5%	15	
Paypal payments	Check out	Accept PayPal payments in check out	Notion	Jira	Early Jun	20%	\$123,000	5%	15	
Geo pricing	Check out	Allow users from major markets (DE, FR, IT,	Notion	Jira	Q3	40%	TBD	0.5%	TBD	
Free trial	Pricing options	Give users 7 days free premium access be	Notion	Jira	Q3	100%	TBD	0.5%	TBD	
Remove ad hoc payments	Pricing options	Remove ability for users to purchase lessor	Notion	Jira	Q3	100%	TBD	0.5%	TBD	
SMS sign up	Account creation	Allow uses to sign up with SMS instead of e	Notion	Jira	Q3	20%	TBD	0.5%	TBD	

<https://www.hustlebadger.com/what-do->

Formulating a task

<action> **the** *<result>* **<by|for|of|to>** *<object>*

Product Backlog

Table 2.9 Code completeness checklist

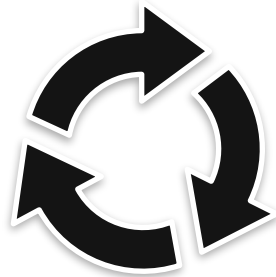
State	Description
Reviewed	The code has been reviewed by another team member who has checked that it meets agreed coding standards, is understandable, includes appropriate comments, and has been refactored if necessary.
Unit tested	All unit tests have been run automatically and all tests have executed successfully.
Integrated	The code has been integrated with the project codebase and no integration errors have been reported.
Integration tested	All integration tests have been run automatically and all tests have been executed successfully.
Accepted	Acceptance tests have been run if appropriate and the Product Owner or the development team has confirmed that the product backlog item has been completed.

Kahoot!

Managing technical risks



People



Process



Technology

Managing technical risks

		Probability		
		Low 1	Medium 2	High 3
Impact	Low 1	1	2	3
	Medium 2	2	4	6
	High 3	3	6	9

A probability/impact matrix for quantifying risk

Actionable points

Identify main functional Software Requirements

Create User Personas

Build Features

Build some User Stories

Build the Backlog

Next time on ...

Finish onboarding Form and GitHub Classroom

Course 3

Please read from *Engineering Software Products* (Sommerville):

- Chapter 4 – Software Architecture (p.92-104)

Please read from *Software Architecture for Developers* (Brown):

- Chapter 4 - What is “software architecture”? (p.6-11)

